

# Current Hybrid Architecture Performance

## Second to Sixth Drawing Sets | Version 10

MODEL\_VERSION: 10.11.24

GATE: P2610E3\_SECOND\_TO\_SIXTH\_FULL\_POPULATION\_LIVE\_VISION\_HYBRID\_ACCURACY\_BENCHMARK\_AND\_RE

PHASE: P2.6.10-E.3

DECISION: PASS\_WITH\_LIMITATIONS

BENCHMARK MODE: LIVE\_BENCHMARK

Confidential / internal use. Shadow benchmark. Not a production promotion.

### 1. EXECUTIVE SUMMARY

Five-set current hybrid architecture: Claude Vision semantics + deterministic engineering.

Model beams: 530 GT beams: 605 Matched: 515

HYBRID: 486 (91.70%) FALLBACK: 44 (8.30%)

Pooled beam identification: 85.12% (515.0 / 605.0)

Pooled bar identification: 48.17% (2008.0 / 4169.0)

Pooled correct-of-detected: 36.50% (733.0 / 2008.0)

Pooled diameter identification: 77.84%

Pooled steel/weight accuracy: 60.64%

Pooled overall accuracy: 57.61%

Model kg: 90305.121 Benchmark kg: 148910.644

### 2. CURRENT HYBRID ARCHITECTURE

DXF -> Beam discovery -> Deterministic geometry context -> Claude Vision

-> D.1 semantic authority contract -> D.2 hybrid resolution

-> D.3 engineering binding -> D.4 shadow engineering calculation

-> accuracy benchmark.

VISION SEMANTICS: target identity, layer, groups, bar count, diameter,

specification, MAIN/EXTRA, support scope, stirrup identification.

DETERMINISTIC ENGINEERING: spacers, geometry, cut length, development,

anchorage, hooks, stirrup engineering, pieces, weight, BBS, workbook.

Vision-preferred is validated, not blindly accepted.

### 3. BENCHMARK SCOPE AND POPULATION

Second, Third, Fourth, Fifth, Sixth. First Set excluded.

Second: model=65 GT=67 matched=64 truth=ESTIMATOR\_EXCEL

Third: model=61 GT=63 matched=57 truth=ESTIMATOR\_EXCEL

Fourth: model=118 GT=143 matched=112 truth=ESTIMATOR\_EXCEL

Fifth: model=143 GT=187 matched=143 truth=ESTIMATOR\_EXCEL

Sixth: model=143 GT=145 matched=139 truth=ESTIMATOR\_EXCEL

### 4. VISION EXECUTION AND HYBRID COVERAGE

Second: eligible=63 attempted=63 new=63 reused=0 retried=0 api=63 schema=63 usable=63 HYBRID=63 FALLBACK=2

Third: eligible=61 attempted=61 new=61 reused=0 retried=0 api=61 schema=61 usable=61 HYBRID=61 FALLBACK=0

Fourth: eligible=76 attempted=76 new=76 reused=0 retried=0 api=76 schema=76 usable=76 HYBRID=76 FALLBACK=42

Sixth: eligible=143 attempted=143 new=143 reused=0 retried=0 api=143 schema=143 usable=143 HYBRID=143 FALLBACK=0  
Fifth Set reuse decision: VISION\_REUSED\_CURRENT\_ARCHITECTURE

## 5. ACCURACY BY DRAWING SET

Second: beam=95.52% bar=65.28% correct=31.35% diameter=78.97% steel=87.95% overall=70.03%  
Third: beam=90.48% bar=54.29% correct=27.27% diameter=74.31% steel=68.25% overall=60.07%  
Fourth: beam=78.32% bar=40.43% correct=41.32% diameter=75.26% steel=54.29% overall=53.59%  
Fifth: beam=76.47% bar=39.29% correct=36.89% diameter=76.75% steel=47.58% overall=50.06%  
Sixth: beam=95.86% bar=59.83% correct=39.38% diameter=81.85% steel=83.18% overall=69.56%

## 6. POOLED SECOND-TO-SIXTH PERFORMANCE

Headline KPIs pool raw numerators and denominators. Steel kg is pooled before accuracy.  
Set percentages are not averaged for the headline overall score.  
Overall = mean(beam ID, bar ID, correct-of-detected, steel/weight). Diameter excluded.

## 7. SEMANTIC ERROR PROFILE

MISSING: 2161  
MATCH: 733  
WRONG\_QUANTITY: 593  
WRONG\_DIAMETER: 445  
EXTRA: 183  
WRONG\_ROLE: 134  
PARTIAL\_MATCH: 103  
ACCEPTABLE\_EXTRA: 89

## 8. ENGINEERING CALCULATION PROFILE

CUT\_LENGTH\_DERIVED\_FALLBACK: 188  
CUT\_LENGTH\_UNAVAILABLE: 0  
STIRRUP\_ENGINEERING\_UNAVAILABLE: 69  
PARTIAL\_CALCULATION: 82  
INCOMPATIBLE: 0  
SPACER\_ZERO: 242

## 9. HYBRID / FALLBACK COHORTS (pooled applicable subsets)

HYBRID\_ONLY: beam=97.52% bar=57.03% correct=36.39% steel=75.22% overall=66.54%  
FALLBACK\_ONLY: beam=97.73% bar=48.43% correct=38.21% steel=71.00% overall=63.84%  
FULL\_POPULATION: beam=85.12% bar=48.17% correct=36.50% steel=60.64% overall=57.61%

## 10. CURRENT WORKFLOW VALUE

Model-assisted estimation. Estimator verification required.  
Accuracy is not equivalent to time saved. No time study in this phase.  
Do not treat Vision API success as engineering accuracy.

## 11. METHODOLOGY, SOURCES AND LIMITATIONS

Truth: estimator Excel (evaluation-only). QA.2A BeamMatcher / BarMatcher / metric8.  
QA.3.0 four-KPI overall. Ground truth is never used in runtime semantic resolution.  
Limitations: ESTIMATOR\_WORKBOOK\_MAPPING\_LIMITATION, FIRST\_SET\_EXCLUDED, SHADOW\_ONLY\_NOT\_PRODUCED  
PRODUCTION\_WRITE=false ENGINEERING\_CHANGES=NONE shadow\_only=true

New live: 343 Reused: 143 Retried: 0 Failed: 0  
Input tokens: 1088060 Output tokens: 213497 Runtime s: 3854.94

## CONCLUSION

This benchmark reports current hybrid architecture performance on Second through Sixth sets. It is not a historical improvement claim. No historical improvement claim. No production readiness claim.